

Question 1: Linear Regression with Mean Absolute Error (MAE)

Guess and Check (inspect plotted points to find a candidate line achieving $\mathcal{L}_{\text{MAE}} < 1.0$).
Given the 4 training samples below, fit a linear model $\hat{y} = wx + b$ that minimizes the Mean Absolute Error loss:

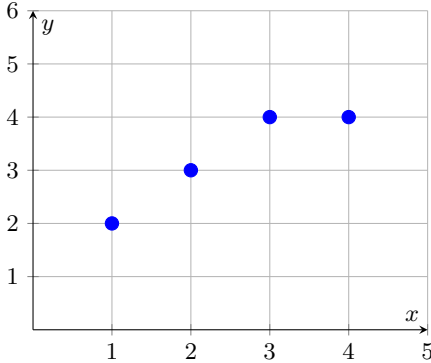
$$\mathcal{L}_{\text{MAE}}(w, b) = \frac{1}{N} \sum_{i=1}^N |(wx_i + b) - y_i|$$

(Note: If multiple optimal lines achieve the same minimum MAE loss, select the line that interpolates the maximum number of training points.)

Training Dataset:

Sample (i)	Feature (x_i)	Target (y_i)
1	1.0	2.0
2	2.0	3.0
3	3.0	4.0
4	4.0	4.0

Plot data points and sketch the optimal line:



Tasks & Questions:

a) Find the optimal parameters (w^*, b^*) that minimize MAE.

$w^* = \rule{1.5cm}{0.4pt}$ $b^* = \rule{1.5cm}{0.4pt}$

b) Calculate the minimum total MAE loss ($\mathcal{L}_{\text{MAE}}^*$).

$\mathcal{L}_{\text{MAE}}^* = \rule{2.5cm}{0.4pt}$

c) Compute predictions $\hat{y}$ for the test points:

- $x = 1.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$
- $x = 2.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$
- $x = 3.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$

Question 2: Linear Regression with Mean Squared Error (MSE)

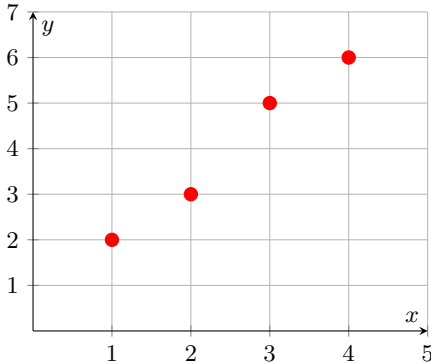
Closed-Form Solution / Normal Equations (derive optimal w^*, b^* analytically using sample means $\bar{x}, \bar{y}$).
Given the 4 training samples below, fit a linear model $\hat{y} = wx + b$ that minimizes the Mean Squared Error loss:

$$\mathcal{L}_{\text{MSE}}(w, b) = \frac{1}{2N} \sum_{i=1}^N ((wx_i + b) - y_i)^2$$

Training Dataset:

Sample (i)	Feature (x_i)	Target (y_i)
1	1.0	2.0
2	2.0	3.0
3	3.0	5.0
4	4.0	6.0

Plot data points and sketch the optimal line:



Tasks & Questions:

a) Derive the optimal weights (w^*, b^*) using analytical formulas $(\bar{x}, \bar{y})$.

$w^* = \rule{1.5cm}{0.4pt}$ $b^* = \rule{1.5cm}{0.4pt}$

b) Calculate the minimum total MSE loss ($\mathcal{L}_{\text{MSE}}^*$).

$\mathcal{L}_{\text{MSE}}^* = \rule{2.5cm}{0.4pt}$

c) Compute predictions $\hat{y}$ for the test points:

- $x = 1.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$
- $x = 2.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$
- $x = 3.5 \implies \hat{y} = \rule{1.5cm}{0.4pt}$

Question 3: Binary Logistic Regression

Decision Boundary & Sigmoid Probability Inference (evaluate $z = 0$ half-spaces, $\hat{p} = \sigma(z)$, and BCE loss). Consider a binary classification task with 2D feature vector $\mathbf{x} = [x_1, x_2]^T$, logit equation $z = w_0 + w_1x_1 + w_2x_2$, and predicted probability $\hat{p} = \sigma(z) = \frac{1}{1+e^{-z}}$. The decision boundary is defined by $x_1 + x_2 = 5$ with optimal parameters $\mathbf{w}^* = [w_0^*, w_1^*, w_2^*]^T = [5.0, -1.0, -1.0]^T$.

Training Dataset:

Sample (i)	x_1	x_2	Class (y_i)
1	1.0	2.0	1
2	2.0	1.0	1
3	3.0	4.0	0
4	4.0	3.0	0

Tasks & Questions:

- a) Sketch the decision boundary $x_1 + x_2 = 5$ on the plot and label regions where $\hat{y} = 1$ vs $\hat{y} = 0$.
- b) Calculate the total Binary Cross-Entropy (BCE) loss under parameter set $\mathbf{w}^*$:

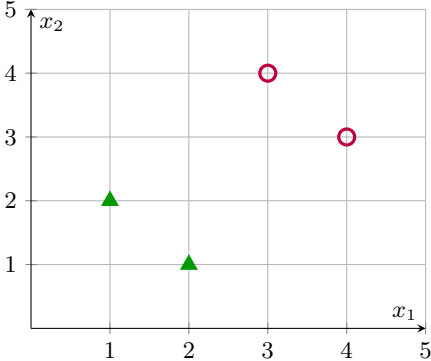
$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \ln \hat{p}_i + (1 - y_i) \ln(1 - \hat{p}_i)]$$

$\mathcal{L}_{\text{BCE}}^* =$ _____

- c) Evaluate predictions for test points (Threshold = 0.5; predict $\hat{y} = 1$ if $\hat{p} \geq 0.5$, else 0):

- $Q_1(1.5, 1.5) \implies \hat{p} =$ _____, $\hat{y} =$ _____
- $Q_2(2.5, 2.5) \implies \hat{p} =$ _____, $\hat{y} =$ _____
- $Q_3(3.5, 3.5) \implies \hat{p} =$ _____, $\hat{y} =$ _____

Plot points and decision boundary:



Question 4: k -Nearest Neighbors Classification

Distance Metrics & Majority Voting (compute Euclidean L_2 and Manhattan L_1 distances with majority voting). Below is a dataset of 10 samples belonging to Class A ($\triangle$) and Class B ($\square$), along with two query points $Q_1(2, 3)$ and $Q_2(4, 4)$ marked with stars ($\star$). (Note: In case of distance or voting ties, break ties by selecting Class A / lower sample ID.)

Training Samples:

ID	Coord	Class	ID	Coord	Class
P_1	(1, 1)	A	P_6	(5, 5)	B
P_2	(1, 3)	A	P_7	(6, 3)	B
P_3	(2, 2)	A	P_8	(7, 5)	B
P_4	(3, 1)	A	P_9	(5, 7)	B
P_5	(3, 3)	A	P_{10}	(7, 7)	B

Tasks & Questions:

- a) For $Q_1(2, 3)$, find the 3 nearest neighbors using **Euclidean distance** $d_E = \sqrt{\Delta x_1^2 + \Delta x_2^2}$:
Neighbors: {_____} $\implies \hat{y}(Q_1) =$ _____
- b) For $Q_2(4, 4)$, classify using **Euclidean distance**:
- $k = 1$: Neighbor = _____ $\implies \hat{y} =$ _____
 - $k = 3$: Neighbors = _____ $\implies \hat{y} =$ _____
- c) For $Q_2(4, 4)$, classify using **Manhattan distance** $d_M = |\Delta x_1| + |\Delta x_2|$ with $k = 3$:
Neighbors: {_____} $\implies \hat{y}(Q_2) =$ _____

